

EX. NO: 2

IMPLEMENTATION OF MONOALPHABETIC CIPHER

AIM:

To write a C program to implement the Monoalphabetic Substitution Cipher technique.

ALGORITHM:

1. Start the program.
2. Enter the plain text.
3. Define a substitution key containing 26 letters.
4. Read each character of the plain text.
5. Replace each alphabet with its corresponding letter in the substitution key.
6. Keep spaces and special characters unchanged.
7. Display the encrypted text.
8. Stop the program.

PROGRAM:

```
#include <stdio.h>
```

```
#include <string.h>
```

```
#include <ctype.h>
```

```
void main()
```

```
{    char plain[100],
```

```
    cipher[100];
```

```
    char key[26] = "QWERTYUIOPASDFGHJKLZXCVBNM";
```

```
    int i;
```

```

    printf("Enter the plain text: ");

fgets(plain, sizeof(plain), stdin);

for(i = 0; plain[i] != '\0'; i++)
{
    if(isupper(plain[i]))
cipher[i] = key[plain[i] - 'A'];

    else if(islower(plain[i]))
cipher[i] = key[plain[i] - 'a'];

    else
        cipher[i]
= plain[i];

}

cipher[i] = '\0';

printf("\nPLAIN TEXT: %s", plain);

printf("ENCRYPTED TEXT: %s", cipher);

}

```

OUTPUT:

```
main.c
15 {
16     if(isupper(plain[i]))
17         cipher[i] = key[plain[i] - 'A'];
18
19     else if(islower(plain[i]))
20         cipher[i] = key[plain[i] - 'a'];
21
22     else
23         cipher[i] = plain[i];
24 }
25
26 cipher[i] = '\0';
27
28 printf("\nPLAIN TEXT: %s", plain);
29 printf("ENCRYPTED TEXT: %s", cipher);
30 }
```

Enter the plain text: MEET ME TOMORROW

PLAIN TEXT: MEET ME TOMORROW

ENCRYPTED TEXT: DTTZ DT ZGDGKKGV

=== Code Exited With Errors ===

RESULT:

Thus, the Monoalphabetic Substitution Cipher was successfully implemented using C language, and the given plaintext was encrypted using the predefined substitution key.